homework10

P325-习题 12.1	A 类	1-1, 1-3, 1-5;
P332-习题 12.2	A 类	1-3, 1-5, 1-7; 2-1, 2-5;
	B 类	1-1, 1-3; 4.

1-1、求级数 $\sum_{n\geq 1}\frac{(-1)^{n-1}}{2^{n-1}}=\frac{1}{1-\frac{-1}{2}}=\frac{2}{3}$

1-3、求级数 $\sum_{n\geq 1}\frac{1}{4n^2-1}$

解:

$$u_n=\frac{1}{4n^2-1}=\frac{1}{2}\left(\frac{1}{2n-1}-\frac{1}{2n+1}\right)$$

$$\Rightarrow S_n=\sum_{k=1}^{n}u_k=\frac{1}{2}\sum_{k=1}^{n}\left(\frac{1}{2k-1}-\frac{1}{2k+1}\right)=\frac{1}{2}\left(1-\frac{1}{2n+1}\right)\to\frac{1}{2}.$$

1-5、求级数 $\sum_{n\geq 1}\frac{n}{(n+1)!}$

解:

$$u_n=\frac{n}{(n+1)!}=\frac{1}{n!}-\frac{1}{(n+1)!}$$

$$\Rightarrow S_n=\sum_{k=1}^{n}u_k=\sum_{k=1}^{n}\left(\frac{1}{k!}-\frac{1}{(k+1)!}\right)=1-\frac{1}{(n+1)!}\to 1$$

1-3、判断敛散性 $\sum_{n\geq 1}(\sqrt[n]{a}-1)^p \quad (a>1)$

解: 记 $u_n=(\sqrt[n]{a}-1)^p$, 则

$$u_n=(e^{\frac{1}{n}\ln a}-1)^p\sim(\ln a)^p\,n^{-p}\Rightarrow \text{级数收敛当且仅当 } p>1$$

1-5、判断敛散性 $\sum_{n\geq 1}\dfrac{1}{2^n-n}$

解: 由 $n/2^n\to 0$ 知

$$u_n=\frac{1}{2^n-n}\sim 2^{-n}\Rightarrow \text{级数收敛}$$

1-7、判断敛散性 $\sum_{n\geq 2}\dfrac{1}{(\ln n)^2}$

解: 记 $u_n=\frac{1}{(\ln n)^{-2}}$, $v_n=n^{-1}$, 则

$$\frac{u_n}{v_n}=\frac{n}{(\ln n)^2}\to+\infty,\quad \sum_{n\geq 1}v_n\text{发散}\Rightarrow \text{级数发散}$$

2-1、判定敛散性 $\sum_{n\geq 1}\frac{1}{4^n}\left(1+\frac{1}{n}\right)^{n^2}$

解:

$$\sqrt[n]{x_n}=\frac{1}{4}\left(1+\frac{1}{n}\right)^{n}\to\frac{e}{4}<1\Rightarrow \text{级数收敛}$$

2-5、判定敛散性 $\sum_{n\geq 1}(\cos\frac{1}{n})^{n^3}$

解:

$$\sqrt[n]{y_n}=\left(\cos\frac{1}{n}\right)^{n^2}=\left(1+(\cos\frac{1}{n}-1)\right)^{n^2}$$

$$n^2\left(\cos\frac{1}{n}-1\right)\to\frac{-1}{2}\Rightarrow\sqrt[n]{y_n}\to e^{-1/2}<1\Rightarrow \text{级数收敛}$$

1-1、判定敛散性 $\sum\limits_{n\geq 1}\int_0^{\frac{1}{n}}\frac{\sqrt{x}}{1+x^4}\,dx$

解: 记 $x_n=\int_0^{\frac{1}{n}}\frac{\sqrt{x}}{1+x^4}\,dx$

$$\Rightarrow x_n\leq\int_0^{\frac{1}{n}}\sqrt{x}\,dx=\frac{2}{3}x^{3/2}\Big|_0^{\frac{1}{n}}=\frac{2}{3}n^{\frac{-3}{2}}\Rightarrow\sum_{n\geq 1}x_n\ \text{收敛}$$

1-3、判定敛散性 $\sum\limits_{n\geq 1}\left(\frac{1}{n^p}-\sin\frac{1}{n^p}\right)$

解: 记 $u_n=\frac{1}{n^p}-\sin\frac{1}{n^p}$

$$p<0\text{ 时}\Rightarrow u_n\geq n^{-p}-1\to+\infty\quad\text{级数发散}$$

$$p=0\text{ 时}\Rightarrow u_n=1-\sin 1\not\to 0\quad\text{级数发散}$$

$$p>0\text{ 时, 由}x-\sin x\sim\frac{1}{6}x^3\Rightarrow x_n\sim\frac{1}{6}\frac{1}{n^{3p}}$$

$$\Rightarrow\text{级数收敛当且仅当 }p>\frac{1}{3}$$

4、设 $a_n > 0$, $S_n = a_1 + a_2 + \cdots + a_n$, 级数 $\sum_{n\geq 1} a_n$ 发散, 证明:

1. $\sum_{n\geq 1} \frac{a_n}{S_n}$ 发散;

反证法, 假设级数收敛, 则 $\frac{a_n}{S_n} = \frac{S_n - S_{n-1}}{S_n} \to 0$

$$\Rightarrow \ln \frac{S_n}{S_{n-1}} = -\ln \frac{S_{n-1}}{S_n} \sim -\frac{S_{n-1} - S_n}{S_n} \sim \frac{a_n}{S_n} \Rightarrow \sum_{n\geq 1} \ln\left(\frac{S_n}{S_{n-1}}\right) \text{也收敛}$$

然而, 由 S_n 发散知 $\sum_{k=2}^{n} \ln\left(\frac{S_k}{S_{k-1}}\right) = \ln \frac{S_n}{S_1} \to +\infty$ 矛盾 $\Rightarrow \sum_{n\geq 1} \frac{a_n}{S_n}$ 发散.

4、设 $a_n > 0$, $S_n = a_1 + a_2 + \cdots + a_n$, 级数 $\sum_{n\geq 1} a_n$ 发散, 证明:

2. $\sum_{n\geq 1} \frac{a_n}{S_n^2}$ 收敛.

$$\text{由 } a_n > 0 \Rightarrow S_n \geq S_{n-1} \Rightarrow \frac{a_n}{S_n^2} = \frac{S_n - S_{n-1}}{S_n^2} \leq \frac{S_n - S_{n-1}}{S_n S_{n-1}} = \frac{1}{S_{n-1}} - \frac{1}{S_n}$$

$$\Rightarrow \text{部分和} \sum_{k=1}^{n} \frac{a_k}{S_k^2} = \frac{a_1}{S_1^2} + \sum_{k=2}^{n} \frac{S_k - S_{k-1}}{S_k^2} \leq \frac{a_1}{S_1^2} + \sum_{k=2}^{n} \left(\frac{1}{S_{k-1}} - \frac{1}{S_k}\right)$$

$$\Rightarrow \text{部分和} \leq \frac{1}{a_1} + \frac{1}{S_1} - \frac{1}{S_n} \leq \frac{2}{a_1} \text{ 有界} \Rightarrow \sum_{n\geq 1} \frac{a_n}{S_n^2} \text{ 收敛.}$$